

Faculty of Applied Science and Technology

LAB #1 - Basic Matlab Application

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Procedures and results:

Add the number of all questions in this report, even if you decided to not answer them.

For each question, you should explain your observation + code snippet.

1. I Opened the Matlab, Software. With previous experience with the software I adjusted the windows and work areas to my liking.
2. After entering the number 190 Matlab displays the exact number under a variable named ans which is short for Answer.
3. After entering the simple expression $100+12$ Matlab rewrites the ans variable from the previous step and gives me 112, which is $100+12$ in Math.
4. Matlab has contained the num1 variable to -1.
5. Matlab has contained the num2 variable to 90.
6. After doing the expression $\text{num1}/\text{num2}$ Matlab displays the ans to be -0.0111 as our Answer to the expression, which if entered on a calculator is correct.
7. After entering more combinations and containers and using operators to see what happens to Matlab I have below:

```
num1 = -1
num2 = 90
num3 = 2015
denom = num1 * num2
denom = -90
num3_over_denom = num3 / denom
num3_over_denom = -22.3889
num4 = num3_over_denom + num3
num4 = 1.9926e+03
```

```

ratio = num2 / num4
ratio = 0.0452
num5 = ratio * 4
num5 = 0.1807
num6 = num5 + 0.02
num6 = 0.2007
final_result = num6 - 0.0007
final_result = 0.2000

```

8. Matlab Textbook Problems, Section 1.10 – Questions 1-10

A) Questions 1-5, Calculate:

1.
 - a) $Q1a = (5 - (19/7) + (2.5)^{(3)})^{(2)}$
 $Q1a = 320.7937$
 - b) $Q1b = 7 * 3.1 + (\text{sqrt}(120)/5) - 15^{(5/3)}$
 $Q1b = -67.3421$
2.
 - a) $Q2a = \text{nthroot}(8 + (80/2.6), 3) + \exp(3.5)$
 $Q2a = 36.5000$
 - b) $Q2b = ((1/(\text{sqrt}(75))) + (73/(3.1)^3))^{(1/4)} + 55 * 0.41$
 $Q2b = 23.8156$
3.
 - a) $Q3a = (23 + \text{nthroot}(45, 3)) / (16 * 0.7) + \log_{10}(589006)$
 $Q3a = 8.1413$
 - b) $Q3b = (36.1 - 2.25 * \pi) * (\exp(2.3) + \text{sqrt}(20))$
 $Q3b = 419.3971$
4.
 - a) $Q4a = ((3.8)^2 / (2.75 - 41 * 25)) + ((5.2 + 1.8^5) / \text{sqrt}(3.5))$
 $Q4a = 12.8656$
 - b) $Q4b = (2.1 * (10^6) - 15.2 * (10^5)) / (3 * (\text{nthroot}(6 * 10^{11}, 3)))$
 $Q4b = 22.9222$
5.
 - a) $Q5a = (\sin(0.2 * \pi)) / (\cos(\pi/6)) + \tand(72)$
 $Q5a = 3.7564$
 - b) $Q5b = ((\tand(64) * \cosd(15))^2) + ((\text{sind}(37)^2) / (\cosd(20)^2))$
 $Q5b = 4.3323$

B) Questions 6-9, Define & Evaluate:

6. Defined the variable z as $z = 4.5$; then evaluated:
 - a) $Q6a = (0.4 * z)^4 + (3.1 * z)^2 - 162.3 * z - 80.7$
 $Q6a = -605.9499$
 - b) $Q6b = (z^3 - 23) / (\text{nthroot}(z^2 + 17.5, 3))$
 $Q6b = 20.3080$

7. Defined the variable t as t = 3.2; then evaluated:

a) $Q7a = ((1/2)*\exp(2*t))-3.81*t^3$

$Q7a = 176.0764$

b) $Q7b = (6*t^2+6*t-2)/(t^2-1)$

$Q7b = 8.5108$

8. Defined the variables x and y as x = 6.5 and y = 3.8; then evaluated:

a) $Q8a = ((x^2+y^2)^(2/3))+((x*y)/(y-x))$

$Q8a = 5.6091$

b) $Q8b = (\text{sqrt}(x+y)/(x-y)^2) + (2*x^2) - (x*y^2)$

$Q8b = -8.9198$

9. Defined the variables a = cd², b=(c+a)/(c-d), c=4.6, and d=1.7; then evaluated:

a) $Q9a = \exp(d-b) + \text{nthroot}(c+a,3) - (c*a)^d$

$Q9a = -1.0861e+03$

b) $Q9b = (d/c) + ((c*t)/b)^2 - c^d - a/b$

$Q9b = -9.4810$

C) Question 10, Trigonometric Identities:

10. For each part, verify that the identity is correct by calculating the values of the left and right sides of the equation, substituting $x = \pi/10$.

a) $Q10a1 = \cos(x)^2 - \sin(x)^2$

$Q10a1 = 8.09$

$Q10a2 = 1-2*\sin(x)^2$

$Q10a2 = 8.09$

b) $Q10b1 = \tan(x)/(\sin(x)-2*\tan(x))$

$Q10b1 = -0.9533$

$Q10b2 = 1/(\cos(x)-2)$

$Q10b2 = -0.9533$

9. Matlab Textbook Problems, Section 1.10 – Question 21 & 25.

A) Question 21:

$a=18;b=7;$

$P = \pi*(a+b)*(3 - (\text{sqrt}((3*a+b)*(a+3*b)))/(a+b))$

$P = 82.3883$ in

B) Question 25:

$V=15; r1=119.8; r2=120.5; r3=121.2; r4=119.3;$

$c = r2/r1; d=r3/r4;$

$Vab = V * ((c-d)/((c+1)*(d+1)))$

$Vab = -0.0374$

10. Matlab Closed. After understanding more about matlabs calculations I am confident in the progress of understanding Mathematics.

11. Tracing Exercise:

eqn	aa = 130 + 4;	Bb = 5/10+2-4; 0.5+2-4 0.5+(-2)	cc = 5 % 2 * 11;
ans	aa = 134	bb = -1.5	Cc = 5 (because % is a comment in matlab)

x = aa - (bb * cc);

x = 134 - (-1.5*5) -> 134 - (-7.5)

x = 141.5

eqn	m1 = 2 ^ 4 - 10; 16-10	k1 = 3 ^ (5 - 11); 3^(-6)	f1 = 87 - 2 / 20; 87-0.1
ans	m1 = 6	K1 = 0.00137	F1 = 86.9

y = m1 * k1 - 10 - f1 / 3 ;

y = 6*0.00137-10-86.9/3 -> 6*0.00137-10-28.967 -> 0.00822-10-28.967 -> -9.99178-28.967

y = -38.95878

eqn	Dd = 4.5 * 2.3 - 10.3; -> 10.35-10.3	Ee = 5.3 + 6.7 - 0.9; -> 5.3+5.8
ans	Dd = 0.05	Ee = 11.1

j = abs (ee) + round (ee)*9 - fix (ee)*10;

j= 11.1+11*9-11*10

j=11.1+99-110

j=11.1+(-11)

j=0.1

z = ceil (ee) - sqrt (dd*dd) * floor (ee) - j;

z = 12 - $\sqrt{(0.05*0.05)}$ *11 - 0.1

z= 12 - 0.05 * 11 - 0.1

z= 12-0.55-0.1

z=11.45-0.1

z=11.35

Correct Answers:

6a->> z=4.5;

>> % Part (a)

>> 0.4*z^4+3.1*z^2-162.3*z-80.7

ans = -584.2500

9b- >> % Part (b)

>> d/c+(c/b)^2-c^d-a/b

ans = -14.6163

10a-10- >> x=pi/10;

>> % Part (a)

>> Left=cos(x)^2-sin(x)^2

Left = 0.8090

>> Right=1-2*sin(x)^2

Right = 0.8090

References:

MATLAB LOGIN: Matlab & Simulink. MATLAB Login | MATLAB & Simulink. (n.d.).

<https://matlab.mathworks.com/>

Gilat, A. (2017). MATLAB: An introduction with applications (6th ed.). John Wiley & Sons.